Voice Activated Home Automation System For Bedridden Patients

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This project can make life easier for bedridden patients and older people as their voice command can trigger the control section that adjusts their bed's position. It can even control home appliances like lights, fan, AC, and TV, based on the command. Even children can operate the lights at night using voice commands without their parents' assistance.

There can be an option for text commands instead of voice commands. The proposed system is user-friendly, and no technical assistance is needed for its operation. The author's prototype is shown in Fig. 1.

A wireless headset worn by a person transmits the person's voice to a smartphone that uses Google Assistant app. The app has voice recognition software for interpreting and translating the voice command into digital data that is understandable by the microcontroller to do the required task(s). The control section includes microcontroller with integrated Wi-Fi, relay, and associated circuitry.

Circuit and working

Block diagram of the system is shown in Fig. 2 and functional block diagram of the developed system is shown in Fig. 3. The system receives voice commands through Google Assistant running on the smartphone. If the command is valid (for instance, Light On), the 'If This, Then That' (IFTTT) applet will send the web service request to Blynk server and convert it into digital data, understandable by the microcontroller, and transfer the data to microcontroller wirelessly through Wi-Fi and actuate the device as per the voice command.

As shown in the circuit diagram in Fig. 4, the system uses an ESP32 board (Board1), which has a microcontroller integrated with Wi-Fi transceiver. When it gets voice command, it enables the relay driver circuitry. In this example, 5V relays are used to switch on/off light, fan, and TV. (A 3-channel relay module may be used in place of individual relays.)

Whenever a valid command is received by ESP32, the program triggers a high or low signal to its GPIO pins connected with relays, which control the home appliances. The servo motor is used to change the bed's back-rest position up or down.

Software implementation

The software, along with Google Assistant app, involves programming of ESP32 using Arduino IDE, configuring Blynk server, and IFTTT applets in the smartphone.

The Blynk app can be configured using the following steps:

1. Download Blynk app from Play Store or App Store.

2. Open the app and create an account. If you have already created an account, you may log in.

3. Start by creating a new project.

4. Create a project name (say, Bedridden_IOT) and choose device as 'ESP32 Dev Board' and connection type as 'WiFi'.

5. Press Create button. A window will pop up which says "Auth Token was sent to: your email." Tap Ok button to proceed. You can open your email to check your authentication key. This key will be used in your Arduino code and IFTTT configurations later on.

6. Tap anywhere on the canvas to open the widget box. All the available widgets are located here.

7. Tap on the particular widget to change the settings. Select the Button widget and rename it as Light and digital output as gp2 and change Mode to Switch option. Similarly, select another Button widget and name

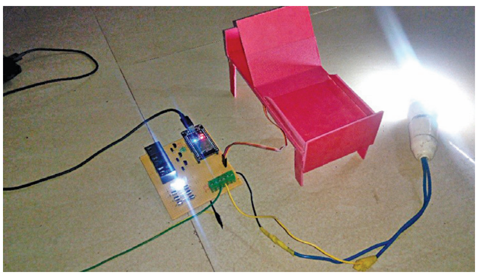

Fig. 1: Author's prototype

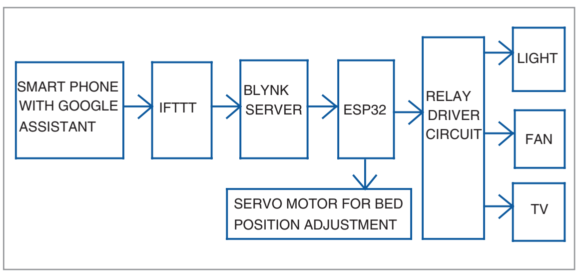

Fig. 3: Functional block diagram

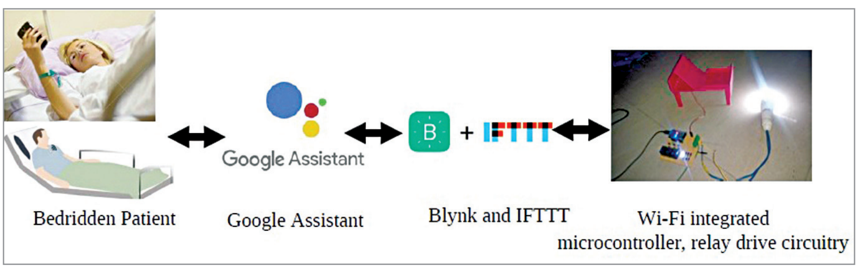

Fig. 2: Block diagram